

## Modeling with a Parent Function: the Open Box

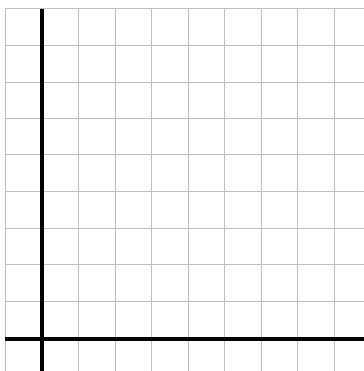
**TEACHER KEY** | Algebra II | Unit 1 | Day 9 | TEKS A2.2.A, A2.7.I

**Objective:** I can build a function that models a real situation, decide which domain makes sense for the context, and use a table or graph to find a maximum.

### The Problem

Start with a flat sheet of cardboard **20 inches by 10 inches**. Cut a square of side  **$x$**  out of each of the four corners, then fold the flaps up to make an open-top box.

Sketch the sheet with the corner squares marked, and label one side  $x$ .



### Part 1 — Build the Function

Each cut removes  $x$  from BOTH ends of a side, so each side loses  $2x$ .

Height of the box:  **$x$**

Length of the box:  **$20 - 2x$**       Width of the box:  **$10 - 2x$**

Volume  $V(x)$  = length  $\cdot$  width  $\cdot$  height =  **$x(20 - 2x)(10 - 2x)$**

Multiplied out:  $V(x) = 4x^3 - 60x^2 + 200x$       Which parent function family? **cubic**

### Part 2 — Find the Domain That Makes Sense

The equation accepts any real number. The cardboard does not.

Why must  $x$  be greater than 0? **you cannot cut a corner of zero or negative size**

The short side is 10 inches. What happens if  $x = 5$ ? **the width  $10 - 2x$  becomes 0, so there is no box**

So the contextual domain is  **$0 < x < 5$**  interval  **$(0, 5)$**  Open or closed? **open - both ends excluded**

### Part 3 — Make a Table

Compute  $V(x) = x(20 - 2x)(10 - 2x)$  for each value. Round to the nearest tenth.

| $x$    | 0.5  | 1   | 1.5   | 2   | 2.5   | 3   | 4  | 4.5  |
|--------|------|-----|-------|-----|-------|-----|----|------|
| $V(x)$ | 85.5 | 144 | 178.5 | 192 | 187.5 | 168 | 96 | 49.5 |

Between which two  $x$ -values does the volume stop growing and start shrinking? **between 2 and 2.5**

Largest volume in your table:  **$192 \text{ in}^3$  at  $x = 2$**  Is that certainly the true maximum? **no - only the best value tested**

## Part 4 — Refine the Maximum

Test values between 2 and 2.5 to close in.

|      |        |        |        |        |        |
|------|--------|--------|--------|--------|--------|
| x    | 2.0    | 2.1    | 2.11   | 2.2    | 2.3    |
| V(x) | 192.00 | 192.44 | 192.45 | 192.19 | 191.36 |

Maximum volume  $\sim 192.45 \text{ in}^3$  when  $x \sim 2.11 \text{ in}$

Box dimensions then: **15.78 by 5.78 by 2.11 in**

Range of the function on this domain:  **$0 < V \leq 192.45$**  interval  **$(0, 192.45]$**

## Part 5 — Interpret in Context

1. What does  $V(1) = 144$  mean in words? **cutting 1-inch corners gives a box holding  $144 \text{ in}^3$**
2. Why is  $x = 5$  excluded but  $x = 4.5$  allowed? **at 5 the width is 0; at 4.5 a thin box still exists**
3. Would a domain of all reals make sense here? **no - only  $0 < x < 5$  describes a real box**

## Practice — A Second Box

A sheet is 16 in. by 8 in. Squares of side  $x$  are cut from the corners. Show your work.

1. Write the volume function.

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$$V(x) = x(16 - 2x)(8 - 2x)$$

2. Which side limits the domain, and what is the limit? **the 8-inch side, so  $x < 4$**
3. Contextual domain:  **$0 < x < 4$ ,  $(0, 4)$**
4. Compute  $V(1)$  and  $V(2)$ . Show your arithmetic.

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$$V(1) = 84 \text{ in}^3 \quad V(2) = 96 \text{ in}^3$$

5. Explain why  $V(4) = 0$  even though 4 is a positive number.

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## Exit Ticket

A sheet is 12 in. by 12 in. Squares of side  $x$  are cut from the corners and the sides folded up.

Volume function:  **$V(x) = x(12 - 2x)^2$**  Contextual domain:  **$0 < x < 6$ ,  $(0, 6)$**

Explain why the upper limit is 6: **at  $x = 6$  both sides become 0, leaving no box**